

Backend Concepts Guide (Project Walkthrough)

1) What is Backend Development?

Backend handles:

- Business logic (rules)
- Data storage/retrieval
- Authentication/authorization
- API contracts for frontend/mobile clients

You can think of backend as "the brain + memory" of an application.

2) Project Architecture Used Here

We use a layered pattern:

1. **Routes**: URL + HTTP method mapping
2. **Controllers**: HTTP-level handling ('req', 'res')
3. **Services**: Business rules and workflows
4. **Repositories**: Data access and persistence details

Why this matters:

- Easier to read and maintain
- Easier to test
- Easier to swap tools (e.g. file DB -> PostgreSQL)

Alternative options:

- MVC-only style (simpler, but can become messy as project grows)
- Clean Architecture / Hexagonal (more scalable but more complex)

3) Authentication (Signed Token)

Flow:

1. User logs in with email/password.
2. Backend verifies credentials.
3. Backend returns signed token.
4. Client sends token in 'Authorization' header.
5. Backend verifies token before protected routes.

Why signed token?

- Stateless
- Common pattern across frontend/mobile/microservices

Other options:

- Server sessions + cookies
- OAuth (Google/GitHub login)

When to use:

- Token-based auth is great for API-based apps.
- Session cookies are good for traditional server-rendered apps.